Validating AI-generated content

We let AI generate content for us based on some textual prompt, but how do we know whether the content, based on patterns in the training data, is actually true—or reasonable enough for our use cases? In fact, much of the content generated by AI could be classified as false, inaccurate, or even nonexistent. It's as if the AI is constructing an alternative reality and presenting it as factual information, known as *hallucinating*. Ensuring the accuracy and validity of AI-generated content not only enhances the user experience but also prevents the spread of misinformation that could lead to poor decision making.

In this book, we will take a hands-on approach to validate our generated content. We will explore techniques such as setting clear objectives, carefully framing the context of responses, adjusting certain LLM parameters, conducting cross-validations, and incorporating robust validation processes.

Handling bias

Bias and discrimination are among the most prevalent problems associated with generative AI technologies, and instances of inaccurate or misleading image generation are of significant concern because they can influence human-made decisions that rely on visual evidence. While there are no easy solutions to tackling bias when using LLMs and generative AI, we can make decisions that will help us in the long run. Restricting the source knowledge base, choosing models trained on diverse datasets, and including bias auditing can help mitigate bias to some extent. *Bias auditing* involves systematically testing a model with diverse inputs to identify and measure any biased behavior.

In this book, we will carefully confine our context and scope of applications to work on factual knowledge bases that do not discriminate against individuals. We will explore practical ways to conduct these audits as we build our applications.

Enhancing user experience

Users interacting with your generative AI web apps expect a high-quality experience that keeps them engaged. This means fast and accurate responses, an intuitive interface, a clear presentation of information, and personalization. User experience and satisfaction are pivotal, especially when adding conversational features. Significant progress has been made in LLMs, enabling them to accept a broader range of media inputs and making them more accessible. These models can now generate videos from text (text-to-video) and more. Since natural language can be ambiguous, we aim to provide users with options beyond text to achieve their goals.

This book prioritizes user experience. We'll discuss techniques such as streaming responses, backpropagation, using accessible components from the Vercel AI SDK, and using multiple formats to reduce frustration from repeated trial and error.